
Financial Dashboard with Chatbot Integration

This project showcases a **financial dashboard** integrated with a **chatbot**.

The main goal is to **represent the financial status of a company** through a bar graph and to provide **interactive chatbot support** that delivers insights based on an inserted Excel sheet.

✅ Tools and Technologies Used:

- **AWS EC2**: Cloud infrastructure to run backend logic.
 - **Python**: For data processing and machine learning.
 - **Pandas**: For data manipulation and analysis.
 - **Multinomial Naïve Bayes (ML)**: For question classification.
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1. 👤 User Interaction

- The user initiates a query using **natural language**, specifically about the inserted dataset.
 - The chatbot captures the question and prepares it for processing.
 - This makes the dashboard **interactive and user-friendly**.
 - **Note**: Questions must be related to the dataset only; unrelated queries are not supported.
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2. 💬 Chatbot

- Acts as the interface between the user and the backend.
 - Sends user queries to a **Python service running on AWS EC2**.
 - The Python service:
 - Understands question types (e.g., totals, averages, date-based lookups).
 - Converts questions into a structured format.
 - Delivers the final response in **natural language**.
 - Ensures smooth interaction with the dataset.
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3. ☁️ AWS (EC2)

- **Provides cloud infrastructure** to host the backend services.
- Ensures **scalability, persistence, and reliability**.
- EC2 is used to:

- Host the Python service.
- Run the chatbot and ML logic continuously.
- Enable frontend-backend communication over the internet.

4. Python

Python handles all the **logic, ML, and data processing**.

Feature	What Python Does
re	Cleans and extracts keywords or dates from user input.
datetime	Parses date strings into Python datetime objects.
input() / print()	Manages the chatbot's interactive loop.
def	Defines key functions: preprocess(), forecast_status_for_date(), answer_question()
LinearRegression	Trains a model to predict future revenue based on timeline index.
TfidfVectorizer + MultinomialNB	Converts text to numerical form and classifies intent of the question.
if-else / try-except / for loops	Handles control flow and decision-making.

5. Pandas

Pandas handles **Excel file processing and data analysis**.

Operation	What Pandas Does
pd.read_excel(...)	Loads the Excel file into a DataFrame (df).
df['Date'] = pd.to_datetime(...)	Converts string dates to datetime format.
df.sort_values(...).reset_index(...)	Sorts data by date and resets row index.
df['DayIndex'] = np.arange(...)	Adds a timeline index for prediction modeling.
df[...] filtering	Filters data inside functions like answer_question().
df.mean(), df.sum(), df.idxmax()	Computes summaries like averages, totals, and max records.

6. Dashboard (Frontend)

- **Visualizes trends and predictions** through bar graphs and responses.

- Used by **internal users or admins** to monitor and analyze financial performance.
 - The dashboard displays data accurately because:
 - The Excel data was properly loaded and cleaned.
 - Forecasts and summaries were computed in the backend.
 - The frontend was connected to the backend via the EC2 instance and API routes.
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